



SQL – Group by & Having & Order by

SQL – Group by

SQL – Group by

- ▶ **GROUP BY** statement groups rows that have the same values in one or more columns.
- ▶ It is commonly used to create summaries, such as total sales by region or number of users by age group.

SQL – Group by

- ▶ **Used with the SELECT statement.**
- ▶ **Groups rows after filtering with WHERE.**
- ▶ **Can be combined with aggregate functions like SUM(), COUNT(), AVG(), etc.**
- ▶ **Filter grouped results using the HAVING clause.**
- ▶ **Comes after WHERE but before HAVING and ORDER BY.**

SQL – Group by

Rules

- All non-aggregated columns in the `SELECT` list must appear in the `GROUP BY`.
- You cannot use `WHERE` to filter results based on aggregate values — that's what `HAVING` is for.

SQL – Group by

► Syntax

Select Column1, aggregate_function(column2)

From table_name

Where condition

GROUP BY column1, column2

SQL – Group by

► Example

Student (ID, Name, Department)

Select Department, Count(*) as Student_count

From Student

Group By Department

SQL – Group by

► Example

I D	Name	Department
1	Ahsan Jawed	Computer Science
2	Sara Ahmed	Software Engineering
3	Bilal Khan	Computer Science
4	Fatima Noor	Data Science
5	Ali Raza	Software Engineering
6	Maria Hassan	Data Science

Department	Student_Count
Computer Science	2
Software Engineering	2
Data Science	2

SQL – Group by

StudentName	Year	Department
Ahsan Jawed	Senior	Computer Science
Bilal Khan	Junior	Computer Science
Fatima Noor	Senior	Computer Science
Sara Ahmed	Sophomore	Humanities
Ali Raza	Junior	Humanities
Maria Hassan	Sophomore	Humanities

Select Department,
Year,
Count(*) as
Student_count
From Student
Group By
Department, Year

SQL – Group by

Department	Year	Student_Count
Computer Science	Junior	1
Computer Science	Senior	2
Humanities	Junior	1
Humanities	Sophomore	2

SQL – Having

SQL – Having

- ▶ **HAVING** clause filters results after applying aggregate functions. Unlike **WHERE**, which filters individual rows, **HAVING** works with aggregated results.

SQL – Having

- ▶ **Filters results based on aggregate functions.**
- ▶ **Supports Boolean conditions (AND, OR).**
- ▶ **Applied after aggregate functions in the query.**
- ▶ **Useful for summary-level or aggregated filtering.**

SQL – Having

► Syntax

```
SELECT  
AGGREGATE_FUNCTION(column_name)  
FROM table_name  
HAVING condition;
```

SQL – Having

```
SELECT column_name1, aggregate_function  
(column_name2)  
FROM table_name  
GROUP BY column_name1  
HAVING condition;
```

SQL – Having - Example

Name	Department	Marks
Ahsan Jawed	Computer Science	85
Sara Khan	Computer Science	92
Bilal Ahmed	Information Tech	78
Fatima Ali	Information Tech	88
Hamza Raza	Business Admin	67
Maryam Noor	Business Admin	74
Zain Sheikh	Computer Science	59
Hira Imran	Information Tech	95

SELECT Department,
AVG(Marks) AS AvgMarks
FROM Student
GROUP BY Department;

Department	AvgMarks
Computer Science	78.67
Information Tech	87.00
Business Admin	70.50

SQL – Having - Example

Name	Department	Marks
Ahsan Jawed	Computer Science	85
Sara Khan	Computer Science	92
Bilal Ahmed	Information Tech	78
Fatima Ali	Information Tech	88
Hamza Raza	Business Admin	67
Maryam Noor	Business Admin	74
Zain Sheikh	Computer Science	59
Hira Imran	Information Tech	95

SELECT Department,
AVG(Marks) AS AvgMarks
FROM Student
GROUP BY Department
HAVING AVG(Marks) > 71.0;

Department	AvgMarks
Computer Science	78.67
Information Tech	87.00

SQL – Order by

SQL – Order By

- ▶ **The ORDER BY clause in SQL is used to sort query results based on one or more columns in either ascending (ASC) or descending (DESC) order.**
- ▶ **Whether you are presenting data to users or analyzing large datasets, sorting the results in a structured way is essential.**

SQL – Order By

- ▶ By default, it sorts in ascending order (lowest to highest).
- ▶ To sort in descending order, use the DESC keyword.

SQL – Order By

► Syntax

```
SELECT * FROM table_name ORDER BY  
column_name ASC | DESC;
```

SQL – Order by - Example

Name	Department	Marks
Ahsan Jawed	Computer Science	85
Sara Khan	Computer Science	92
Bilal Ahmed	Information Tech	78
Fatima Ali	Information Tech	88
Hamza Raza	Business Admin	67
Maryam Noor	Business Admin	74
Zain Sheikh	Computer Science	59
Hira Imran	Information Tech	95

**SELECT * FROM
Student ORDER BY
Name;**

SQL – Order By

Name	Department	Marks
Ahsan Jawed	Computer Science	85
Bilal Ahmed	Information Tech	78
Fatima Ali	Information Tech	88
Hamza Raza	Business Admin	67
Hira Imran	Information Tech	95
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Sara Khan	Computer Science	92
Zain Sheikh	Computer Science	59

SQL – Order By

Name	Department	Marks
Zain Sheikh	Computer Science	59
Sara Khan	Computer Science	92
Maryam Noor	Business Admin	74
Hira Imran	Information Tech	95
Hamza Raza	Business Admin	67
Fatima Ali	Information Tech	88
Bilal Ahmed	Information Tech	78
Ahsan Jawed	Computer Science	85

```
SELECT *  
FROM Student  
ORDER BY Name  
DESC;
```



THANK YOU